

CS 459/559 Spring 2026
Capture the Flag Assignment
Deadline: March 26, 2026

In this assignment, the students will learn and practice cyber offensive techniques in the form of a Capture-The-Flag (CTF) contest. The contest is done using the CTF5G platform where all the vulnerabilities are implanted in an emulated 5G network. You can go to this website to learn the basics of 5G network structure:

- [Chapter 3: Basic Architecture — 5G Mobile Networks: A Systems Approach Version 0.1 documentation](#)

Your task in this assignment is to exploit as many vulnerabilities inside the 5G network as possible. Currently there are eight types of vulnerabilities you can exploit inside the CTF5G platform:

1. APT Configuration Exploitation
2. DOM Cross Site Scripting
3. Meet in the Middle Crypto Attack
4. Packet Sniffing
5. Timing Side Channel Attack
6. TCP Syn Flood Attack
7. Return Oriented Programming Attack

Once an attack has been performed successfully, it will be shown on the leaderboard.

The first successful attack earns 40 points, and each additional successful attack earns 10 points.

- The highest possible score: 100
- Bonus points up to 25 can be handed out, if a student can help add a new attack to the testbed.

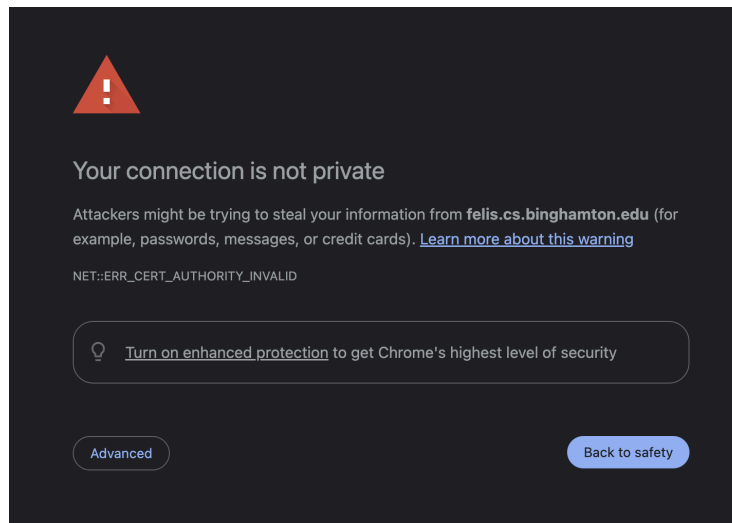
CTF5G Tutorial

1. Click on the following URL to access the Game Dispatcher:

<https://felis.cs.binghamton.edu:4245>

Note:

1. If you're accessing the website off-campus, use the University VPN.
2. If you are using campus WiFi and still unable to access the website, use the University VPN.
3. If the following page shows up:

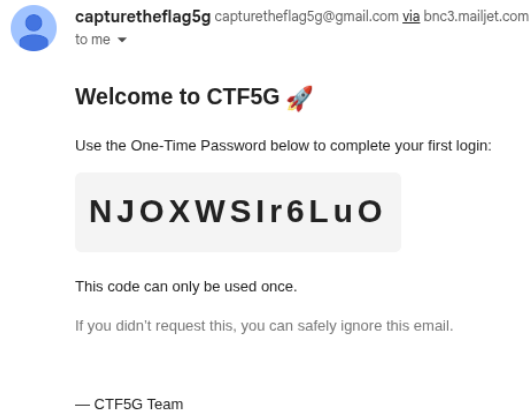


Click on [Advanced](#) and then click on [Proceed to felis.cs.binghamton.edu \(unsafe\)](#)

2. **(If you already have an account, skip to step 4)** Click on the *signup* button to register into CTF5G. Enter the preferred username and password combination, followed by Binghamton email address and click on *signup*.

Note: Use **only alphanumeric** for username, **no special characters**.

3. Upon successful registration, you will be redirected back to the login page(wait for a few seconds). Check your email for an OTP, the email should look like this:



4. For your **initial login**, use your username and the **OTP as the password**. Subsequent logins will require your chosen username and password.
5. The username and the **Slice Id** will be visible at the top left corner of the **Home** screen, as shown in fig 1. Please note this.

DO NOT share your Slice Id with other students.

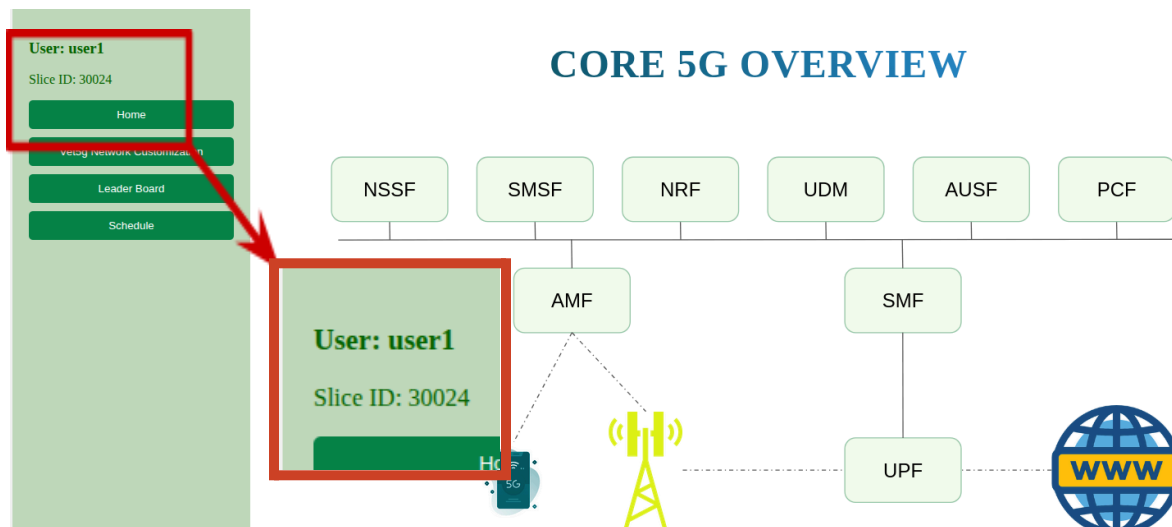


Fig 1: Slice Id within the Home Page

Modules in the Dispatcher:

1. Scheduler
2. Attack Selection
3. LeaderBoard

1. Scheduler

We can only support 15-20 students using the testbed at a time. The scheduler provides an overview of active users and the expected wait time for new users.

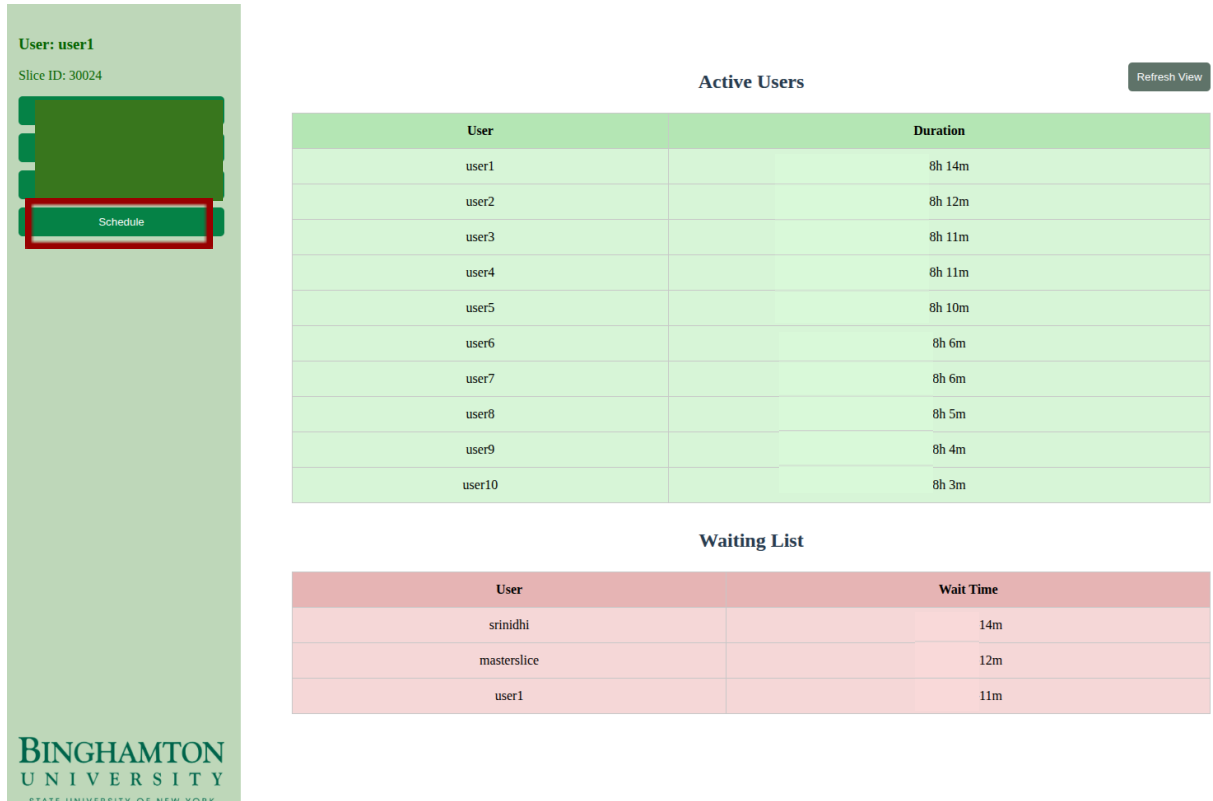


Fig 2: Schedule Layout (**NOT Real Values**)

2. Attack Selection

- 2.1. Next, click on the *Start Capture The Flag* tab on the Home Screen or *Attack Selection* tab to view/select the attacks.
- 2.2. Select your preferred category of attacks, then select the attacks you wish to perform from the sublists.
- 2.3. For more information on each attack, select the attack then click on *Learn More*

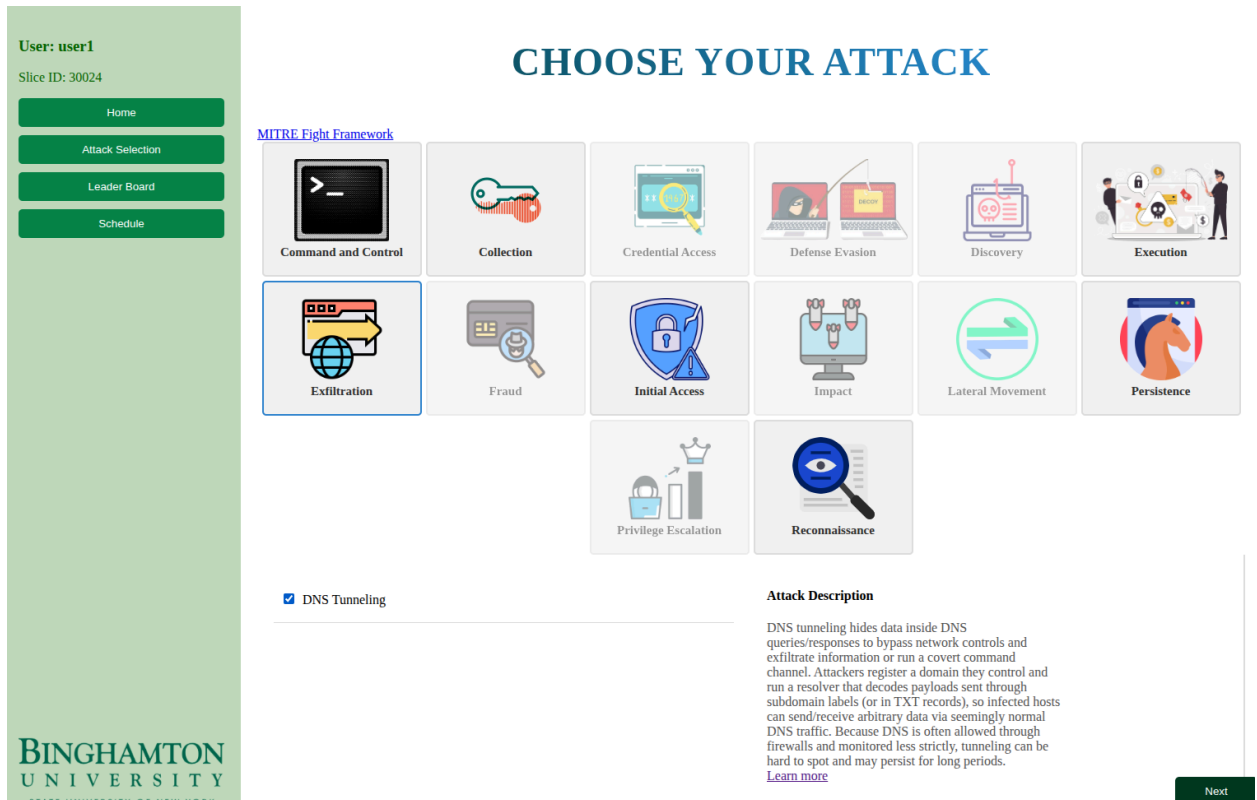


Fig 3: Attack Selection

- 2.4. Only **after selecting** all the preferred attacks click on the *Next* button, in the bottom right.
- 2.5. This opens the *LaunchTestbed* page. Click on the *Start* button to start the testbed. Upon successful launch, 3 buttons will appear.
 - Access Service
 - AMF Logs
 - UE Logs
- 2.6. Click on the *AccessService* button to access the *Ubuntu* container. Click on *AMF Logs* to view the logs of AMF and click on *UE Logs* to view the logs of UE.

Note:

1. Do not Navigate to any other page in CTF5G once you start the testbed, this action will lose the context and you will have to restart the testbed.
2. If the testbed is running for over 4 hours, it will be automatically stopped and you will not be able to restart it.

3. **DO NOT** refresh the terminal page. One accessed by clicking on *AccessService*.
 4. Run `apt update` in the terminal before installing any packages.
- 2.7. Click on the *Stop* button to stop the testbed.

3. LeaderBoard

All successful attacks can be viewed via clicking on the LeaderBoard tab. This page displays the successful attacks of all the participants